Information Theory Intuitively

For B.Tech EC Graduate Who Ratta-Fied It In College
Real-Life Examples Only

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Ye document ek **prequel** hai, Higher-MI walkthrough se pehle padhne ke liye. College me information theory ek chhota chapter tha, jaldbaazi me padhaya gaya, ratta maar ke paper me likh diya gaya — concepts kabhi clear nahi hue. Is document ka maqsad hai: formulas to tumhe pehle se yaad hain, bas unka matlab real-life examples se samjhana. Koi taala-chaabi nahi, koi cryptography nahi — sirf sensors, commute, cricket, aur mausam jaisi roz ki cheezein.

Is version (V28) me ek aur baat theek ki gayi hai, jo khud tumne point out ki thi: is document me pichhli baar bahut jagah **calculation pehle, samjhana baad me** ho gaya tha — jaise formula seedha likh diya, number nikaal diya, khatam. Lekin maqsad calculation karna nahi, **concept clear karna** hai. Isliye ab is document me har calculation se pehle uska **WHY** (kyun nikaal rahe hain) bataya jaayega, phir poora formal sentence ("Given that ..., the probability that ... will be:") likha jaayega, phir formula, aur formula ke **turant baad** uska story-matlab ("Iska matlab ...") likha jaayega. Calculation sirf ek tool hai concept dikhane ka, khud maqsad nahi hai.

Isi version me ek purana sawaal bhi address kiya gaya hai, jo tumne khud uthaya tha: **beech ke calculation me decimal kyun, poora fraction kyun nahi?** Isliye is document me har probability *pehle ek fraction ki tarah* likhi jaayegi — jaise 34/100, kisi din-ginti se nikla hua — aur decimal sirf *aakhri substitution line* par aayega. Isse har number ka source (kitne din me se kitne din) poora padhte waqt dikhta rahega.

How to Read This Document: Notation

Poore document me, yehi symbols baar-baar aayenge. Ek baar yahan dekh lo, phir aage padhte waqt rukna nahi padega.

Symbol	Matlab
X, Y, Z	Random variables (jaise, commute-mode, ya ek sensor ki reading)
$p(x)$	X ke outcome x ki probability, din-ginti se — jaise 34/100
$S(x)$	Surprise ka ek outcome — $-\ln p(x)$, official quantity, entropy se pehle
$H(X)$	Entropy — X ka average confusion, Y jaane se pehle
$H(X \mid Y)$	Conditional entropy — Y jaanne **ke baad** X me **bachi** confusion
$I(X;Y)$	Mutual information — Y jaanne se X ka confusion **kitna kam hua**
$I(X;Y \mid Z)$	Conditional mutual information — Z pehle se pata hone par, reduction
Ω	O-information — redundant (> 0) ya synergistic (< 0)
nats	Entropy ki unit, jab ln (natural log) use karte hain

Paanch cheezein yaad rakhne layak hain, aage ke chapters ke liye:

1. $S(x)$ ek **outcome-level** quantity hai, surprise. $H(X)$ aur $H(X \mid Y)$ dono **average** (final) quantities hain.
2. $I(X;Y)$ ek **reduction** hai — do final quantities ka **farak** ($H(X) - H(X \mid Y)$), khud final nahi hai.
3. Poore document me probability **pehle fraction form** me likhi jaayegi (din-ginti se, jaise 34/100). Decimal sirf jab usi fraction ka numeric substitution ho raha ho, usi line par, pehli baar aayega.

4. Har example ka source batate hain — "itne din me se itne din" — taaki koi number random na lage, grounded lage.

5. Har calculation se **pehle** uska **WHY** bataya jaayega, aur **baad** me uska **story-matlab**. Koi formula akela, bina kisi matlab ke, nahi aayega — shauk samjhane ka hai, sirf formality ki nahi.

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1 Entropy: Confusion, Not Probability

1.1 What Is Entropy

Jammu me October ka ek subah socho. Ghar se nikalne se pehle, exact temperature pata nahi hai. Ye na-maloom hona hi **confusion** hai, ya **uncertainty**.

Thermometer dekhne se pehle, confusion high hai — temperature $10°C$ bhi ho sakta hai, $20°C$ bhi. Thermometer dekhne ke baad, confusion 0 ho jaata hai — ab exact number pata hai.

Entropy isi confusion ko ek number me measure karta hai. Entropy jitna zyada, utna zyada na-maloom.

Ek bilkul concrete tasveer: band dabba

"Confusion" ko aur concrete banane ke liye, ek tasveer pakad lo, aur isi tasveer ko is poore document me baar-baar use karenge.

Socho table par ek **band dabba** hai. Dabbe ke andar coin rakha hai — Heads ya Tails, par tumhe nahi pata kya hai. Dabba band hai, tum andar nahi dekh sakte.

Dabbe ko dekhte hi (ya koi bata de ki andar kya hai), confusion 0 ho jaata hai — tumhe exactly pata chal gaya. Dabba band rehte hue, confusion maximum hai — Heads ya Tails, dono equally possible lagte hain.

Yahi band dabba, is poore document me har jagah use hoga — kabhi coin ke saath, kabhi commute-mode ke saath, kabhi match-score ke saath. Jab bhi "confusion" shabd aaye, bas yaad karo: *dabba band hai, andar kya hai pata nahi.*

1.2 Surprise of One Outcome

Entropy define karne se pehle, ek chhoti quantity define karte hain: **surprise**. Ye entropy ka ek **separate, official** quantity hai — entropy ka koi unofficial by-product nahi.

Agar ek outcome x ki probability $p(x)$ hai, uska surprise:

$$S(x) = -\ln p(x).$$

Rare outcome (chhota $p(x)$) ka surprise zyada hota hai. Common outcome (bada $p(x)$) ka surprise kam hota hai.

Entropy is surprise ka **average** hai, saare outcomes par, unki apni probability se weighted:

$$H(X) = \sum_x p(x)\, S(x) = -\sum_x p(x) \ln p(x).$$

Matlab: entropy "average surprise" hai, poore random variable ka. Weight probability isliye hai kyunki frequent outcome average me zyada baar contribute karta hai — rare outcome kam baar aata hai, isliye uska weight bhi kam hona chahiye.

1.3 Why Logarithm, Not Probability Directly

Yahan ek natural sawaal uthta hai: surprise ke liye $-\ln p(x)$ kyun, seedha $p(x)$ kyun nahi?

Wajah: independent events ka information **add** hona chahiye, jabki unki probability **multiply** hoti hai.

Case: One Coin Versus Two Coins

Setup: Ek fair coin, do outcomes (Heads, Tails), dono ki probability 1/2.

Step 1: One coin ka surprise

$$S = -\ln \frac{1}{2} = \ln 2 \approx 0.693 \text{ nats.}$$

Step 2: Do independent coins ka surprise
Do independent coins ka ek specific outcome (Heads-Heads) ki probability:

$$p(\text{HH}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Iska surprise:

$$S = -\ln \frac{1}{4} = \ln 4 \approx 1.386 \text{ nats.}$$

Step 3: Additivity check karo
Do coins ka surprise, ek coin ke surprise ka double hai:

$$\ln 4 = \ln 2 + \ln 2.$$

Ye additivity sirf logarithm se milti hai, kyunki $\ln(a \times b) = \ln a + \ln b$. Agar surprise seedha probability se define karte, ye additivity nahi milti.

1.4 Why 0.693, Not 0.5

Yahan ek common confusion aati hai: fair coin ki probability 1/2 hai, par entropy $\ln 2 \approx 0.693$ aata hai. Dono alag numbers kyun?

Wajah: 1/2 aur $\ln 2$ do **alag quantities** hain.

- 1/2 — ek specific outcome (Heads) ki **probability**, ek single toss me.
- $\ln 2 \approx 0.693$ — **average surprise**, poore random variable ka. Agar coin ko bahut baar (jaise 1000 baar) toss karo, average surprise $\ln 2$ nats/toss aayega.

Dono alag hain kyunki surprise khud $-\ln p$ hai, p nahi. $p = 1/2$ ke liye surprise $-\ln(1/2) = \ln 2$ hota hai — ye 1/2 nahi, ek alag number hai, logarithm se aata hai.

Case: Fair Coin Entropy

Setup: Fair coin, Heads aur Tails dono ki probability 1/2.

Step 1: Har outcome ka surprise

Heads ka surprise:

$$S(\text{Heads}) = -\ln\frac{1}{2} = \ln 2 \approx 0.693 \text{ nats.}$$

Tails ka surprise bhi same hai, symmetry se:

$$S(\text{Tails}) = -\ln\frac{1}{2} = \ln 2 \approx 0.693 \text{ nats.}$$

Step 2: Weighted average lo

$$H(X) = \frac{1}{2}(\ln 2) + \frac{1}{2}(\ln 2) = \ln 2 \approx 0.693 \text{ nats.}$$

Entropy $\ln 2 \approx 0.693$ hai — do outcomes ka weighted average surprise, probability $1/2$ nahi.

Band-dabba se jodo: coin wala dabba band hai, Heads ya Tails equally possible lagte hain — isliye average surprise maximum ($\ln 2$) hai. Agar coin biased hota (ek outcome common, doosra rare), dabba khulte hi usually wahi common outcome nikalta — average surprise kam hota.

Note: Bits Versus Nats

Is document me entropy ln (natural log) se nikal rahe hain, isliye unit **nats** hai. EC courses me (communication, coding theory) aksar $\log_2$ use hota hai, unit **bits** — fair coin ka entropy wahan 1 bit aata hai, 0.693 nats nahi. Dono sahi hain, sirf logarithm ka base alag hai: $\log_2 x = \ln x / \ln 2$. Is document me hum ln (nats) use karenge, kyunki Higher-MI walkthrough bhi isi convention me hai.

Key Takeaway

- Entropy average surprise hai: $H(X) = \sum_x p(x)\, S(x)$, jahan $S(x) = -\ln p(x)$.
- Surprise aur entropy alag quantities hain — surprise ek outcome ka, entropy poore variable ka average.
- Independent events ka surprise add hota hai, logarithm isi additivity ke liye use hota hai.
- Fair coin ki probability $1/2$ hai, par entropy $\ln 2 \approx 0.693$ hai — dono alag quantities hain.

1.5 A Non-Trivial Entropy Example: Commute Mode

Ab ek aisa example lete hain jisme probabilities bilkul fair-coin jaisi nahi hain — real observation se, din-ginti se nikli hui.

Story: Jonathan Jammu me office ja raha hai. Kabhi Bike leta hai, kabhi Car — depend karta hai mausam, traffic, aur us din ki mood par. Pichhle 100 din ka record dekha gaya: kitne din Bike liya, kitne din Car.

Case: Commute Mode (Bike Versus Car)

Setup: Pichhle 100 din me se, Jonathan ne 53 din Bike liya, 47 din Car liya.

$$p(\text{Bike}) = \frac{53}{100}, \qquad p(\text{Car}) = \frac{47}{100}.$$

Kyun? Entropy $H(X)$ nikalna hai — matlab, bina kuch dekhe, Jonathan ke commute-mode ke baare me average kitna confusion hai, ye number se dikhana hai. Iske liye pehle har outcome ka surprise chahiye, phir unka weighted average.

Step 1: Surprise table

Outcome x	$p(x)$	$S(x) = -\ln p(x)$
Bike	$\frac{53}{100}$	$-\ln \frac{53}{100} \approx 0.6349$
Car	$\frac{47}{100}$	$-\ln \frac{47}{100} \approx 0.7550$

Step 2: Weighted average lo, $H(X)$ milega

$$\begin{aligned} H(X) &= \frac{53}{100}\left[-\ln \frac{53}{100}\right] + \frac{47}{100}\left[-\ln \frac{47}{100}\right] \\ &= \frac{53}{100}(0.6349) + \frac{47}{100}(0.7550) \\ &= 0.3365 + 0.3549 \\ &= 0.6914 \text{ nats.} \end{aligned}$$

Iska matlab: 53/100 aur 47/100 fair-coin ke $1/2, 1/2$ se close hain par barabar nahi — isliye entropy 0.6914 bhi $\ln 2 \approx 0.693$ ke close hai, par thoda kam, kyunki Bike thoda zyada common hai. Yahi 53/100 aur 47/100 wali commute-mode probability aage Chapter 3 ke full worked example me bhi istemal hogi.

2 Conditional Entropy: Remaining Confusion After Knowing Y

2.1 Two Thermometers, Same Room

Socho tumhare room me do thermometers lage hain — Sensor A aur Sensor B, dono same room ka temperature naap rahe hain. X = Sensor A ki reading (High ya Low), Y = Sensor B ki reading.

Sensor B dekhne se *pehle*, Sensor A ke baare me poora confusion hai — room ka temperature High bhi ho sakta hai, Low bhi. Sensor B dekhne ke *baad* (maan lo Y =High), Sensor A ke baare me bahut kam confusion bachta hai — dono sensors same room naap rahe hain, bas chhota measurement-noise ka farak hoga (Sensor A corner me hai, Sensor B window ke paas, jahan dhoop seedha lagti hai).

Isi "Y jaanne ke baad X me bachi confusion" ko **conditional entropy** kehte hain, $H(X \mid Y)$.

2.2 Definition

$$H(X \mid Y) = \sum_y p(y)\, H(X \mid Y = y),$$

jahan har y ke liye:

$$H(X \mid Y = y) = \sum_x p(x \mid y)\big[-\ln p(x \mid y)\big].$$

2.3 Independent Case: Jammu Coin, Delhi Coin

Ab ek doosra example lo jahan koi connection nahi hai. X = Jammu me ek coin-toss ka result. Y = Delhi me ek alag, independent coin-toss ka result.

Delhi ka result (Y) jaanne se Jammu ke result (X) ke baare me kuch bhi extra nahi pata chalta — dono bilkul unrelated hain. Dono fair coins hain, isliye:

$$H(X \mid Y) = H(X) = \ln 2 \approx 0.693 \text{ nats}.$$

Y jaanne se X ka confusion *bilkul kam nahi hua* — jitna pehle tha, utna hi raha.

2.4 Partial Case: Two Home Thermometers (Numeric)

Story: Yehi do thermometers wapas lo — Sensor A (room ke corner me), Sensor B (window ke paas, dhoop ki wajah se thoda garam reading deta hai). Dono ek hi room naap rahe hain, isliye zyada-tar agree karte hain, par placement ka farak hone se kabhi-kabhi disagree bhi karte hain. Pichhle 100 din ka joint record:

Case: Two Home Thermometers — Numeric

Setup: Pichhle 100 din me se:

X (Sensor A)	Y (Sensor B)	$p(x, y)$
High	High	47/100
High	Low	9/100
Low	High	7/100
Low	Low	37/100

Zyada-tar din dono sensors agree karte hain (High-High ya Low-Low, $47/100 + 37/100 = 84/100$ din), kyunki same room naap rahe hain. Kabhi-kabhi placement-noise ki wajah se disagree karte hain ($9/100 + 7/100 = 16/100$ din).

Step 1: Marginal $p(x)$ nikaalo

$$p(X = \text{High}) = \frac{47}{100} + \frac{9}{100} = \frac{56}{100}.$$
$$p(X = \text{Low}) = \frac{7}{100} + \frac{37}{100} = \frac{44}{100}.$$

Step 2: Marginal $p(y)$ nikaalo

$$p(Y = \text{High}) = \frac{47}{100} + \frac{7}{100} = \frac{54}{100}.$$
$$p(Y = \text{Low}) = \frac{9}{100} + \frac{37}{100} = \frac{46}{100}.$$

Kyun? Ab $p(x \mid y)$ chahiye — matlab, Sensor B (Y) dekhne ke **baad** Sensor A (X) ke baare me kya pata chalta hai, ye jaanna hai. Y =High dekhne ke baad X =High kitna likely hai, ye conditional batayega, tabhi hum bachi hui confusion $H(X \mid Y = \text{High})$ nikaal payenge. Formula: $p(x \mid y) = p(x, y)/p(y)$, yani joint ko us y-branch ke total se renormalize karna.

Step 3: Har y ke liye conditional $p(x \mid y)$ nikaalo

Y =High ke liye:

Given that $Y = \text{High}$, the probability that $X = \text{High}$ will be:

$$p(X = \text{High} \mid Y = \text{High}) = \frac{47/100}{54/100} = \frac{47}{54} \approx 0.8704.$$

Iska matlab: 54 din me se, jab Sensor B High tha, un 54 din me se 47 din Sensor A bhi High tha — isliye Y =High dekhne ke baad X =High hone ki probability 87% ho jaati hai, confusion kaafi kam.

Given that $Y = \text{High}$, the probability that $X = \text{Low}$ will be:

$$p(X = \text{Low} \mid Y = \text{High}) = \frac{7/100}{54/100} = \frac{7}{54} \approx 0.1296.$$

Iska matlab: Sirf 13% cases me disagreement hai — Sensor B High par bhi Sensor A Low.

Y =Low ke liye:

Given that $Y = \text{Low}$, the probability that $X = \text{High}$ will be:

$$p(X = \text{High} \mid Y = \text{Low}) = \frac{9/100}{46/100} = \frac{9}{46} \approx 0.1957.$$

Iska matlab: Jab Sensor B Low dikhata hai, 46 din me se sirf 9 din Sensor A High tha — yani 20% disagreement, baaki 80% agreement.

Given that $Y = \text{Low}$, the probability that $X = \text{Low}$ will be:

$$p(X = \text{Low} \mid Y = \text{Low}) = \frac{37/100}{46/100} = \frac{37}{46} \approx 0.8043.$$

Iska matlab: Sensor B Low dikhane par, 80% baar Sensor A bhi Low hota hai — dono zyada-tar saath chalte hain.

Kyun? Ab har branch (Y =High aur Y =Low) ke andar, X me kitna confusion bacha hai, ye nikalna hai. Iske liye har branch ka apna surprise table banate hain, phir us branch ka weighted average $H(X \mid Y = y)$ liya jaata hai.

Step 4: Har y ke liye surprise table, phir $H(X \mid Y = y)$
Y =High branch:

x	$p(x \mid Y = \text{High})$	$S = -\ln p(x \mid Y = \text{High})$
High	$47/54$	≈ 0.1388
Low	$7/54$	≈ 2.0430

$$\begin{aligned} H(X \mid Y = \text{High}) &= \frac{47}{54}(0.1388) + \frac{7}{54}(2.0430) \\ &= 0.1209 + 0.2648 \\ &= 0.3857 \text{ nats.} \end{aligned}$$

Iska matlab: Y =High dekhne ke baad, X me sirf 0.3857 nats confusion bachta hai — Y dekhne se pehle poora $H(X)$ (aage Step 5 ke paas compare karenge) ka confusion tha, ab High-branch me ye kaafi ghat gaya, kyunki 87% cases me X bhi High nikalta hai.

Y =Low branch:

x	$p(x \mid Y = \text{Low})$	$S = -\ln p(x \mid Y = \text{Low})$
High	$9/46$	≈ 1.6312
Low	$37/46$	≈ 0.2179

$$\begin{aligned} H(X \mid Y = \text{Low}) &= \frac{9}{46}(1.6312) + \frac{37}{46}(0.2179) \\ &= 0.3192 + 0.1752 \\ &= 0.4943 \text{ nats.} \end{aligned}$$

Iska matlab: Y =Low branch me X ka bacha hua confusion 0.4943 nats hai — High-branch (0.3857) se thoda zyada, kyunki Low-branch me agreement (80%) thoda kam strong hai High-branch ke agreement (87%) se.

Kyun? Dono branches (Y =High aur Y =Low) ka apna-apna bacha confusion mil gaya. Ab ye dono branch-level confusion ko unki apni probability se weight karke average karenge, tabhi overall $H(X \mid Y)$ milega — matlab, Y jaanne ke baad, average me X me kitna confusion bachta hai.

Step 5: Average lo, $H(X \mid Y)$ milega

$$\begin{aligned} H(X \mid Y) &= \frac{54}{100}(0.3857) + \frac{46}{100}(0.4943) \\ &= 0.2083 + 0.2274 \\ &= 0.4357 \text{ nats.} \end{aligned}$$

Iska matlab: Compare karo: agar sensors bilkul independent hote, $H(X \mid Y)$ poora $H(X)$ jitna hota. Isi example ke marginal $p(X = \text{High}) = 56/100$, $p(X = \text{Low}) = 44/100$ se, bina Y ke, $H(X) \approx 0.6859$ nats aata hai; yahan $H(X \mid Y) \approx 0.4357$ hai — kaafi kam, kyunki Sensor B jaanne se Sensor A ka confusion achha-khaasa ghat gaya.

Key Takeaway

- $H(X \mid Y)$ matlab: Y jaanne ke baad, X me **bachi hui** average confusion.
- Agar X, Y connected hain (same-room sensors), $H(X \mid Y)$ chhota hota hai.
- Agar X, Y independent hain (Jammu-Delhi coins), $H(X \mid Y) = H(X)$ — koi kami nahi.
- Har probability fraction form me rakhi gayi (47/100 jaisi), decimal sirf surprise aur final-average ki substitution line par aaya.

3 Mutual Information: Reduction, Not Final Value

3.1 The Core Confusion to Fix

Pichhle chapter me $H(X \mid Y)$ dekha — Y jaanne *ke baad* X me kitna confusion bachta hai.

Ab ek naya sawaal: Y jaanne se X ka confusion **kitna kam hua**? Ye **mutual information**, $I(X;Y)$, batata hai.

Zaroori farak: $H(X \mid Y)$ ek **final value** hai (kitna bacha). $I(X;Y)$ ek **reduction** hai (kitna kam hua). Dono alag quantities hain, alag numbers de sakte hain.

3.2 Definition

$$I(X;Y) = H(X) - H(X \mid Y).$$

Matlab: *pehle* ka confusion ($H(X)$), minus *baad* ka confusion ($H(X \mid Y)$) — yani kitna kam hua.

Ye symmetric bhi hai:

$$I(X;Y) = H(Y) - H(Y \mid X).$$

3.3 Two Home Thermometers: Reduction Example

Chapter 2 wale thermometer example ko wapas lo:

$$H(X) \approx 0.6859 \text{ nats}, \qquad H(X \mid Y) \approx 0.4357 \text{ nats}.$$

Kyun? Ab pichhle chapter ke $H(X)$ aur $H(X \mid Y)$ ko seedha formula $I(X;Y) = H(X) - H(X \mid Y)$ me daal kar dekhte hain ki Sensor B jaanne se Sensor A ka confusion kitna kam hua.

Reduction:

$$\begin{aligned} I(X;Y) &= H(X) - H(X \mid Y) \\ &= 0.6859 - 0.4357 \\ &= 0.2502 \text{ nats}. \end{aligned}$$

Iska matlab: Sensor B jaanne se, Sensor A ka confusion ≈ 0.2502 nats kam hua.

3.4 Symmetry Check: Same Reduction, Doosri Taraf Se

$I(X;Y)$ symmetric hai: $I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X)$. Isi Chapter-2 ke numeric example par ye verify karte hain.

Case: Two Home Thermometers — Verify Symmetric Formula

Setup: Wahi joint table, Chapter 2 se: $p(\text{High}, \text{High}) = 47/100$, $p(\text{High}, \text{Low}) = 9/100$, $p(\text{Low}, \text{High}) = 7/100$, $p(\text{Low}, \text{Low}) = 37/100$.

Kyun? Pehle $H(Y)$ chahiye — yani bina X dekhe, Sensor B ke baare me akela kitna confusion hai, isi tarah jaise Chapter 1 me $H(X)$ nikala tha.

Step 1: $H(Y)$ **nikaalo**
Marginal $p(Y = \text{High}) = 54/100$, $p(Y = \text{Low}) = 46/100$ (Chapter 2, Step 2 se).

$$\begin{aligned} H(Y) &= \frac{54}{100}\left[-\ln\frac{54}{100}\right] + \frac{46}{100}\left[-\ln\frac{46}{100}\right] \\ &= \frac{54}{100}(0.6162) + \frac{46}{100}(0.7765) \\ &= 0.3327 + 0.3572 \\ &= 0.6899 \text{ nats.} \end{aligned}$$

Kyun? Ab $H(Y \mid X)$ chahiye — yani Sensor A (X) dekhne **ke baad** Sensor B (Y) me bacha confusion. Iske liye pehle $p(y \mid x)$ nikalna hoga, bilkul usi tarah jaise Chapter 2 me $p(x \mid y)$ nikala tha, bas is baar roles ulte hain.

Step 2: $H(Y \mid X)$ **nikaalo**
Given that $X = \text{High}$, the probability that $Y = \text{High}$ will be:

$$p(Y = \text{High} \mid X = \text{High}) = \frac{47/100}{56/100} = \frac{47}{56} \approx 0.8393.$$

Iska matlab: Jab Sensor A High tha, 56 din me se 47 din Sensor B bhi High tha — 84% agreement.

Given that $X = \text{High}$, the probability that $Y = \text{Low}$ will be:

$$p(Y = \text{Low} \mid X = \text{High}) = \frac{9/100}{56/100} = \frac{9}{56} \approx 0.1607.$$

Iska matlab: Sirf 16% cases me, Sensor A High par bhi Sensor B Low nikalta hai.

Given that $X = \text{Low}$, the probability that $Y = \text{High}$ will be:

$$p(Y = \text{High} \mid X = \text{Low}) = \frac{7/100}{44/100} = \frac{7}{44} \approx 0.1591.$$

Iska matlab: Sensor A Low hone par, sirf 16% baar Sensor B High dikhata hai.

Given that $X = \text{Low}$, the probability that $Y = \text{Low}$ will be:

$$p(Y = \text{Low} \mid X = \text{Low}) = \frac{37/100}{44/100} = \frac{37}{44} \approx 0.8409.$$

Iska matlab: Sensor A Low hone par, 84% baar Sensor B bhi Low hai — dono sensors zyada-tar saath chalte hain, chaahe kisi bhi taraf se dekho.

Kyun? Ab X =High aur X =Low, dono branches ka apna surprise table banakar, har branch ka bacha confusion $H(Y \mid X = x)$ nikalte hain.

$$\begin{aligned} H(Y \mid X = \text{High}) &= \frac{47}{56}(0.1753) + \frac{9}{56}(1.8279) \\ &= 0.1472 + 0.2938 \\ &= 0.4410 \text{ nats.} \end{aligned}$$

Iska matlab: Sensor A High ho to, Sensor B ka bacha confusion 0.4410 nats hai.

$$\begin{aligned} H(Y \mid X = \text{Low}) &= \frac{7}{44}(1.8384) + \frac{37}{44}(0.1733) \\ &= 0.2925 + 0.1457 \\ &= 0.4382 \text{ nats.} \end{aligned}$$

Iska matlab: Sensor A Low ho to bhi, Sensor B ka bacha confusion lagbhag same, 0.4382 nats hai — dono branches me almost barabar reduction milta hai.

Kyun? Dono branches ka weighted average lekar overall $H(Y \mid X)$ milega.

$$\begin{aligned} H(Y \mid X) &= \frac{56}{100}(0.4410) + \frac{44}{100}(0.4382) \\ &= 0.2470 + 0.1928 \\ &= 0.4398 \text{ nats.} \end{aligned}$$

Kyun? Ab doosri taraf se $I(X;Y)$ nikalte hain — $H(Y)$ minus $H(Y \mid X)$ — aur dekhte hain ki ye Chapter 2-3 me pehli taraf se nikala $I(X;Y) \approx 0.2502$ ke barabar aata hai ya nahi.

Step 3: $I(X;Y)$ is taraf se nikaalo

$$\begin{aligned} I(X;Y) &= H(Y) - H(Y \mid X) \\ &= 0.6899 - 0.4398 \\ &= 0.2501 \text{ nats.} \end{aligned}$$

Iska matlab: Dono taraf se ≈ 0.250 nats (chhota rounding-farak sirf intermediate decimals ki wajah se). X se Y ka confusion kam karna, aur Y se X ka confusion kam karna — dono me equal reduction milta hai. Isi liye $I(X;Y)$ ko "mutual" (dono-taraf-se-shared) information kehte hain, sirf ek-taraf-ki quantity nahi.

3.5 The Identical Case: Where the Confusion Happens

Ab wahi case lo jo pehle confusion create karta hai: $Y = X$ bilkul (jaise, ek sensor ki reading hi doosri jagah copy ho rahi hai).

Case: Identical Coins ($Y = X$)

Setup: X ek fair coin hai. Y bilkul X ka copy hai — jo bhi X aaye, Y wahi hoga.

Kyun? Pehle dekhte hain Y dekhne se **pehle** X ke baare me kitna confusion hai.

Step 1: $H(X)$ **— pehle ka confusion**
Y dekhne se *pehle*, X ke baare me poora confusion hai, fair coin jitna:

$$H(X) = \ln 2 \approx 0.693 \text{ nats.}$$

Kyun? Ab dekhte hain Y dekhne ke **baad** kitna confusion bachta hai. Yahan $Y = X$ hai, isliye Y dekhte hi X turant pata chal jaayega.

Step 2: $H(X \mid Y)$ **— baad ka confusion**
Y dekhne ke *baad* (aur $Y = X$ hai), X turant pata chal jaata hai. Koi confusion nahi bachta:

$$H(X \mid Y) = 0.$$

Kyun? Ab reduction nikalte hain — Y jaanne se X ka confusion kitna kam hua, $H(X)$ minus $H(X \mid Y)$.

Step 3: $I(X;Y)$ **— kitna kam hua**

$$\begin{aligned} I(X;Y) &= H(X) - H(X \mid Y) \\ &= \ln 2 - 0 \\ &= \ln 2 \\ &\approx 0.693 \text{ nats.} \end{aligned}$$

Iska matlab: $Y = X$ hone ki wajah se, Y jaanne se X ka **poora** confusion ($\ln 2$) khatam ho gaya — maximum possible reduction, kyunki final confusion 0 tak girkar poora reduction de gaya.

3.6 Resolving the Confusion: 0 Versus $\ln 2$

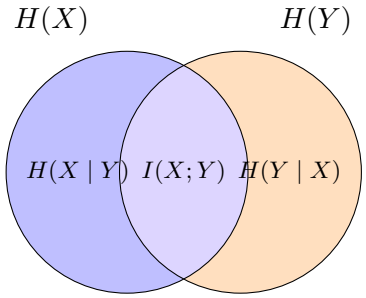
Yahan do alag numbers hain, aur dono sahi hain, kyunki dono alag sawaalon ke jawab hain.

- "Y jaanne ke baad X me kitna confusion *bacha* hai?" — jawab $H(X \mid Y) = 0$. Sahi hai, kyunki $Y = X$ jaante hi X pura pata chal jaata hai.
- "Y jaanne se X ka confusion kitna *kam* hua?" — jawab $I(X;Y) = \ln 2$. Ye bhi sahi hai, kyunki confusion poore $\ln 2$ se girkar 0 par aaya — **poora** confusion kam hua, yani **maximum possible** reduction.

$I(X;Y) = 0$ hone ke liye, Y ko X ka confusion **bilkul na** ghatana padta — yani Y jaanne ke baad bhi utna hi confusion bachta jitna pehle tha. Ye sirf independent case me hota hai (Jammu-Delhi coins, Chapter 2 se), identical case me nahi.

Yaad rakhne wali baat: 0 final uncertainty ka number hai. $\ln 2$ reduction ka number hai. Jab $Y = X$, final uncertainty 0 hai (sab pata chal gaya) *kyunki* reduction poora $\ln 2$ tha (sab confusion kam ho gaya).

3.7 Picture: Two Overlapping Circles



Circle X ka poora area $H(X)$ hai, circle Y ka poora area $H(Y)$ hai. Dono ka **overlap** $I(X;Y)$ hai — jo information dono share karte hain.

X ka akela (non-overlap) hissa $H(X \mid Y)$ hai — jo confusion Y jaanne ke baad bhi X me bachta hai.

Identical case me ($Y = X$), dono circles **poori tarah** overlap karte hain — overlap (I) poora circle jitna bada hai, non-overlap hissa ($H(X \mid Y)$) zero hai.

Key Takeaway

- $H(X \mid Y)$ = final confusion, Y jaanne ke baad bacha hua.
- $I(X;Y)$ = reduction, Y jaanne se kitna kam hua. Formula: $I(X;Y) = H(X) - H(X \mid Y)$.
- $Y = X$ case me: final confusion 0 hai, reduction $\ln 2$ hai. Dono sahi hain, alag sawaalon ke jawab.

3.8 Full Worked Example: Commute Mode and Punctuality

Ab saari cheezein — entropy, joint entropy, conditional entropy, mutual information — ek hi real-life story me ek saath dekhte hain.

Story: Jonathan office jaane ke liye kabhi Bike leta hai, kabhi Car. Uska mode aur uska **on-time pohochna** connected hain — Bike traffic me se nikal jaata hai (kam late), par Car thodi zyada reliable hoti hai jab traffic halka ho. Dono bilkul independent nahi hain, par bilkul identical bhi nahi — ek partial connection hai. Pichhle 100 din ka record: us din kaunsa mode use hua (X), aur us din office on-time pahuncha ya late (Y).

Case: Commute Mode Versus Punctuality — Full Numeric

Setup: Pichhle 100 din me se:

X (Mode)	Y (Punctuality)	$p(x,y)$
Bike	OnTime	34/100
Bike	Late	19/100
Car	OnTime	9/100
Car	Late	38/100

Step 1: Marginal $p(x)$ aur $p(y)$ nikaalo

$$p(\text{Bike}) = \frac{34}{100} + \frac{19}{100} = \frac{53}{100}. \qquad p(\text{Car}) = \frac{9}{100} + \frac{38}{100} = \frac{47}{100}.$$

$$p(\text{OnTime}) = \frac{34}{100} + \frac{9}{100} = \frac{43}{100}. \qquad p(\text{Late}) = \frac{19}{100} + \frac{38}{100} = \frac{57}{100}.$$

Step 2: Surprise table — marginals

Outcome	p	$S = -\ln p$
Bike	53/100	≈ 0.6349
Car	47/100	≈ 0.7550
OnTime	43/100	≈ 0.8440
Late	57/100	≈ 0.5621

Kyun? Chapter 1 jaisa hi kadam — bina ek-doosre ko dekhe, X akela kitna confusion hai aur Y akela kitna confusion hai, ye $H(X)$ aur $H(Y)$ nikaal kar dekhte hain. Ye baseline hai, jisse aage compare karenge.

$$
\begin{aligned}
H(X) &= \frac{53}{100}(0.6349) + \frac{47}{100}(0.7550) \\
&= 0.3365 + 0.3549 \\
&= 0.6914 \text{ nats.}
\end{aligned}
$$

$$
\begin{aligned}
H(Y) &= \frac{43}{100}(0.8440) + \frac{57}{100}(0.5621) \\
&= 0.3629 + 0.3204 \\
&= 0.6833 \text{ nats.}
\end{aligned}
$$

Iska matlab: Mode akela dekha jaaye to 0.6914 nats confusion hai, punctuality akela dekhi jaaye to 0.6833 nats — dono fair-coin ($\ln 2 \approx 0.693$) ke kaafi close hain, kyunki dono roughly 53-47 aur 43-57 split hain.

Kyun? Ab X aur Y ko **saath** dekhna hai — yani dono ka joint outcome (x, y) kitna confusion rakhta hai, ye $H(X, Y)$ (joint entropy) batayega. Ye is poore chapter ki building-block hai, kyunki isi se aage $H(X \mid Y)$ nikalenge, seedha conditional ki jagah.

Step 3: Surprise table — joint outcomes

(x, y)	$p(x, y)$	$S = -\ln p(x, y)$
Bike,OnTime	34/100	≈ 1.0788
Bike,Late	19/100	≈ 1.6607
Car,OnTime	9/100	≈ 2.4079
Car,Late	38/100	≈ 0.9676

Step 4: Joint entropy $H(X, Y)$, ek outcome per line

$$
\begin{aligned}
H(X,Y) &= \frac{34}{100}\left[-\ln\frac{34}{100}\right] + \frac{19}{100}\left[-\ln\frac{19}{100}\right] + \frac{9}{100}\left[-\ln\frac{9}{100}\right] + \frac{38}{100}\left[-\ln\frac{38}{100}\right] \\
&= \frac{34}{100}(1.0788) + \frac{19}{100}(1.6607) + \frac{9}{100}(2.4079) + \frac{38}{100}(0.9676) \\
&= 0.3668 + 0.3155 + 0.2167 + 0.3677 \\
&= 1.2667 \text{ nats.}
\end{aligned}
$$

Iska matlab: 1.2667 nats hai mode **aur** punctuality, dono ek saath, kitna total confusion rakhte hain — jaahir hai ye $H(X)$ ya $H(Y)$ akele se zyada hai, kyunki do variables ka joint confusion dono ka milakar hota hai.

Kyun? Ab $H(X \mid Y)$ chahiye — yani punctuality jaanne ke baad mode me kitna confusion bachta hai. Ise seedha $p(x \mid y)$ se nikalne ki jagah, ek shortcut identity use karte hain: $H(X, Y) = H(Y) + H(X \mid Y)$ — matlab, joint confusion barabar hai (akela Y ka confusion) plus (Y jaanne ke baad X ka bacha confusion). Ye **chain rule** kehlata hai, aur iska story-matlab ye hai: pehle Y ko jaano, phir Y ke baad bacha X ka confusion jod do — total joint confusion mil jaata hai. Isi identity ko ulta karke $H(X \mid Y)$ nikaal sakte hain, bina naya $p(x \mid y)$ table banaye.

Step 5: Conditional entropy, chain rule se

Conditional entropy direct nikalne ke bajaye, identity $H(X,Y) = H(Y) + H(X \mid Y)$ use karte hain:

$$\begin{aligned} H(X \mid Y) &= H(X,Y) - H(Y) \\ &= 1.2667 - 0.6833 \\ &= 0.5834 \text{ nats.} \end{aligned}$$

Iska matlab: Punctuality (Y) jaanne ke baad, mode (X) ka bacha confusion 0.5834 nats hai — $H(X) = 0.6914$ se kam, isliye punctuality se mode ke baare me kuch pata chalta hai, par poora nahi (identical case jaisa 0 nahi aaya).

Symmetric taraf se:

$$\begin{aligned} H(Y \mid X) &= H(X,Y) - H(X) \\ &= 1.2667 - 0.6914 \\ &= 0.5753 \text{ nats.} \end{aligned}$$

Iska matlab: Mode (X) jaanne ke baad, punctuality (Y) ka bacha confusion 0.5753 nats hai — isi tarah $H(Y) = 0.6833$ se kam.

Kyun? Ab mutual information nikalte hain — mode aur punctuality ke beech kitna shared information hai, do tareeke se check karenge: pehla joint-entropy identity se, doosra seedha $H(X) - H(X \mid Y)$ se.

Step 6: Mutual information $I(X;Y)$

$$\begin{aligned} I(X;Y) &= H(X) + H(Y) - H(X,Y) \\ &= 0.6914 + 0.6833 - 1.2667 \\ &= 0.1080 \text{ nats.} \end{aligned}$$

Check, doosre formula se:

$$\begin{aligned} I(X;Y) &= H(X) - H(X \mid Y) \\ &= 0.6914 - 0.5834 \\ &= 0.1080 \text{ nats.} \end{aligned}$$

Iska matlab: $I(X;Y) \approx 0.1080$ nats — na zero (independent) hai, na $\ln 2$ (identical) hai. Mode jaanne se, punctuality ke baare me thoda confusion kam hota hai — jaisa real zindagi me hota hai: commute mode ek factor hai, par akela factor nahi (traffic ka din-waar mood, meetings ka time, aur bhi cheezein punctuality decide karti hain).

4 Three Cases: Independent, Identical, Partial

4.1 Overview

Teen cases hain jo $I(X;Y)$ le sakta hai: zero (independent), maximum (identical), ya beech mein kahin (partial). Teeno real-life se dekhte hain.

Case: Independent — Jammu and Chennai Weather

X = Jammu ka aaj ka temperature-bucket (High/Low). Y = Chennai ka aaj ka temperature-bucket (High/Low). Dono sheher bahut door hain, mausam-system alag hai.

Chennai ka temperature jaanne se Jammu ke baare me kuch nahi pata chalta — dono weather systems unconnected hain. Ye definition se $I(X;Y) = 0$ ka case hai — Venn diagram me dono circles bilkul alag hote hain, koi overlap nahi.

Case: Identical — Sensor and Its Exact Copy

X = Sensor A ki reading. Y = Sensor A ki hi reading, ek doosri jagah (jaise display par) **exact copy** ki gayi.

Y bilkul X jaisa hai, isliye Y jaanne se X **pura** pata chal jaata hai: $I(X;Y) = H(X)$ (maximum possible). Fair-coin-jaisi reading ke liye, ye $\ln 2 \approx 0.693$ hoga — bilkul Chapter 3 wale identical-coin case jaisa. Venn diagram me dono circles **poori tarah** overlap karte hain.

Case: Partial — Two Sensors in the Same Room, One Noisy

X = Sensor A (room ke corner me). Y = Sensor B (room ke window ke paas, thoda noisy — dhoop seedha lagti hai).

Dono same room naap rahe hain, isliye kuch shared information hai. Par Sensor B ka apna extra noise bhi hai, isliye perfect match nahi: $0 < I(X;Y) < H(X)$.

Chapter 2 ka numeric example ($H(X \mid Y) \approx 0.4357$, $I(X;Y) \approx 0.2502$) isi case ka hai.

4.2 Quick Gallery: More Partial-Overlap Examples

Partial-overlap case ($0 < I(X;Y) < H(X)$) sabse common hai real life me. Kuch aur examples, jo aane wale chapters me bhi kaam aayenge:

- **Commute mode aur punctuality**: X =Bike/Car, Y =OnTime/Late, same din ka. Mode kuch reduction deta hai (kuch overlap), par meetings ka time, traffic ka din-waar mood bhi alag factors hain (kuch non-overlap). Isliye $0 < I(X;Y) < H(X)$.
- **Cricket match, pehli aur doosri innings**: X =Team A ka pehli innings score-bucket (High/Low), Y =Team A ka doosri innings score-bucket. Pitch aur team-form thodi carry hoti hai innings se innings (kuch shared information), par fresh bowling-conditions aur match-pressure bhi alag hote hain har innings me (kuch non-shared). Isliye $0 < I(X;Y) < H(X)$.
- **Do mobile phones ka signal strength**: X =Phone 1 ka signal-bucket (Strong/Weak), Y =Phone 2 ka signal-bucket, dono ek hi room me, same network par. Dono ek hi tower se signal le rahe hain (kuch shared information), par har phone ka apna antenna-position aur hardware thoda alag hota hai (kuch non-shared noise). Isliye $0 < I(X;Y) < H(X)$.

Teeno examples me, do variables *related* hain par *identical nahi* — yahi partial-overlap case ki pehchaan hai.

4.3 Numeric Verification: Cricket Innings

Story: Team A pehli innings me ek score banata hai, phir doosri innings khelta hai. Pitch dheere-dheere degrade hoti hai aur team ka momentum carry hota hai, isliye dono innings ka score-bucket *related* hota hai — par fresh bowling-figures aur match-pressure har innings me alag bhi hote hain, isliye perfect match nahi. Pichhle 100 matches ka record:

Case: First Innings Score Versus Second Innings Score

Setup: Pichhle 100 matches me se, Team A ka score-bucket (≥ 160 =High, < 160 =Low), dono innings:

X (1st innings)	Y (2nd innings)	$p(x, y)$
High	High	48/100
High	Low	7/100
Low	High	5/100
Low	Low	40/100

Pitch aur team-form thodi carry hoti hai (High-High aur Low-Low zyada common, $48/100 + 40/100 = 88/100$), par fresh bowling-figures aur match-pressure har innings alag bhi hote hain (High-Low aur Low-High bhi possible, $7/100 + 5/100 = 12/100$).

Step 1: Marginal $p(x), p(y)$ aur surprise table

$$p(X = \text{High}) = \frac{48}{100} + \frac{7}{100} = \frac{55}{100}, \qquad p(X = \text{Low}) = \frac{5}{100} + \frac{40}{100} = \frac{45}{100}.$$
$$p(Y = \text{High}) = \frac{48}{100} + \frac{5}{100} = \frac{53}{100}, \qquad p(Y = \text{Low}) = \frac{7}{100} + \frac{40}{100} = \frac{47}{100}.$$

Outcome	p	$S = -\ln p$
X =High	55/100	≈ 0.5978
X =Low	45/100	≈ 0.7985
Y =High	53/100	≈ 0.6349
Y =Low	47/100	≈ 0.7550

$$H(X) = \frac{55}{100}(0.5978) + \frac{45}{100}(0.7985) = 0.3288 + 0.3593 = 0.6881 \text{ nats.}$$
$$H(Y) = \frac{53}{100}(0.6349) + \frac{47}{100}(0.7550) = 0.3365 + 0.3549 = 0.6914 \text{ nats.}$$

Iska matlab: Dono innings akele dekhne par 0.6881 aur 0.6914 nats confusion hai, dono roughly fair-coin ke close — ab dekhte hain ki ek innings ka score jaanne se doosri innings ke baare me kuch pata chalta hai ya nahi.

Kyun? Ab $p(x \mid y)$ chahiye — matlab, doosri innings (Y) ka score dekhne **ke baad** pehli innings (X) ke baare me kya pata chalta hai. Ye conditional hi aage $H(X \mid Y = y)$ ka base banega.

Step 2: $p(x \mid y)$ nikaalo, har y ke liye

Y =High ke liye:

Given that $Y = \text{High}$, the probability that $X = \text{High}$ will be:

$$p(X = \text{High} \mid Y = \text{High}) = \frac{48/100}{53/100} = \frac{48}{53} \approx 0.9057.$$

Iska matlab: Jin 53 matches me doosri innings High thi, unme se 48 match me pehli innings bhi High thi — 91% agreement, pitch aur momentum-carry ki wajah se.

Given that $Y = \text{High}$, the probability that $X = \text{Low}$ will be:

$$p(X = \text{Low} \mid Y = \text{High}) = \frac{5/100}{53/100} = \frac{5}{53} \approx 0.0943.$$

Iska matlab: Sirf 9% cases me, doosri innings High hone par bhi pehli innings Low thi.

Y =Low ke liye:

Given that $Y = \text{Low}$, the probability that $X = \text{High}$ will be:

$$p(X = \text{High} \mid Y = \text{Low}) = \frac{7/100}{47/100} = \frac{7}{47} \approx 0.1489.$$

Iska matlab: Doosri innings Low hone par, sirf 15% baar pehli innings High thi.

Given that $Y = \text{Low}$, the probability that $X = \text{Low}$ will be:

$$p(X = \text{Low} \mid Y = \text{Low}) = \frac{40/100}{47/100} = \frac{40}{47} \approx 0.8511.$$

Iska matlab: Doosri innings Low hone par, 85% baar pehli innings bhi Low thi — dono innings ka score zyada-tar saath chalta hai.

Kyun? Ab har branch (Y =High, Y =Low) ke andar pehli innings (X) ka bacha confusion $H(X \mid Y = y)$ nikalte hain — iske liye har branch ka apna surprise table banate hain.

Step 3: Har y ke liye surprise table, phir $H(X \mid Y = y)$

$x \mid Y$ =High	p	$S = -\ln p$	$x \mid Y$ =Low	p	$S = -\ln p$
High	48/53	≈ 0.0991	High	7/47	≈ 1.9042
Low	5/53	≈ 2.3609	Low	40/47	≈ 0.1613

$$H(X \mid Y = \text{High}) = \frac{48}{53}(0.0991) + \frac{5}{53}(2.3609) = 0.0898 + 0.2227 = 0.3125 \text{ nats.}$$

$$H(X \mid Y = \text{Low}) = \frac{7}{47}(1.9042) + \frac{40}{47}(0.1613) = 0.2836 + 0.1373 = 0.4209 \text{ nats.}$$

Iska matlab: Jab doosri innings High thi, pehli innings ka bacha confusion sirf 0.3125 nats hai — bahut kam, kyunki 91% agreement strong signal hai. Jab doosri innings Low thi, bacha confusion thoda zyada, 0.4209 nats, kyunki Low-branch ka agreement (85%) thoda weaker hai.

Step 4: Average lo

$$H(X \mid Y) = \frac{53}{100}(0.3125) + \frac{47}{100}(0.4209) = 0.1656 + 0.1978 = 0.3634 \text{ nats.}$$

Kyun? Ab reduction nikalte hain — doosri innings ka score jaanne se, pehli innings ke baare me confusion kitna kam hua.

Step 5: $I(X;Y)$ nikaalo

$$I(X;Y) = H(X) - H(X \mid Y) = 0.6881 - 0.3634 = 0.3247 \text{ nats.}$$

Iska matlab: Pehli innings ka score jaanne se, doosri innings ke baare me confusion 0.3247 nats kam hota hai — kaafi badi reduction, pitch aur momentum-carry ki wajah se.

4.4 Numeric Verification: Mobile Signal Strength

Story: Ek hi room me do phones hain, dono ek hi network tower se signal le rahe hain — isliye zyada-tar ek saath Strong ya ek saath Weak dikhayenge. Par har phone ka apna antenna-position aur hardware thoda alag hota hai, isliye kabhi-kabhi disagree bhi karte hain. Pichhle 100 measurement-instances ka record:

Case: Two Phones, Same Room, Same Tower

Setup: Pichhle 100 instances me se:

X (Phone 1)	Y (Phone 2)	$p(x,y)$
Strong	Strong	35/100
Strong	Weak	10/100
Weak	Strong	9/100
Weak	Weak	46/100

Ek hi tower se signal aa raha hai, isliye zyada-tar dono phones agree karte hain ($35/100 + 46/100 = 81/100$). Har phone ka apna antenna-position aur hardware thoda alag hai, isliye kabhi-kabhi disagree bhi karte hain ($10/100 + 9/100 = 19/100$).

Step 1: Marginal aur surprise table

$$p(X = \text{Strong}) = \frac{45}{100}, \qquad p(X = \text{Weak}) = \frac{55}{100}.$$
$$p(Y = \text{Strong}) = \frac{44}{100}, \qquad p(Y = \text{Weak}) = \frac{56}{100}.$$

Outcome	p	$S = -\ln p$
X =Strong	$45/100$	≈ 0.7985
X =Weak	$55/100$	≈ 0.5978
Y =Strong	$44/100$	≈ 0.8210
Y =Weak	$56/100$	≈ 0.5798

$$H(X) = \frac{45}{100}(0.7985) + \frac{55}{100}(0.5978) = 0.3593 + 0.3288 = 0.6881 \text{ nats.}$$
$$H(Y) = \frac{44}{100}(0.8210) + \frac{56}{100}(0.5798) = 0.3612 + 0.3247 = 0.6859 \text{ nats.}$$

Kyun? Ab Phone 2 (Y) dekhne ke baad Phone 1 (X) ke baare me kya pata chalta hai, ye $p(x \mid y)$ se dekhte hain.

Step 2: $p(x \mid y)$ **nikaalo**
Given that $Y = \text{Strong}$, the probability that $X = \text{Strong}$ will be:

$$p(X = \text{Strong} \mid Y = \text{Strong}) = \frac{35/100}{44/100} = \frac{35}{44} \approx 0.7955.$$

Iska matlab: Jab Phone 2 Strong dikhata hai, 80% baar Phone 1 bhi Strong hota hai — ek hi tower se signal aane ka asar.

Given that $Y = \text{Strong}$, the probability that $X = \text{Weak}$ will be:

$$p(X = \text{Weak} \mid Y = \text{Strong}) = \frac{9/100}{44/100} = \frac{9}{44} \approx 0.2045.$$

Iska matlab: Sirf 20% cases me, Phone 2 Strong par Phone 1 Weak nikalta hai — apne antenna-position ki wajah se.

Given that $Y = \text{Weak}$, the probability that $X = \text{Strong}$ will be:

$$p(X = \text{Strong} \mid Y = \text{Weak}) = \frac{10/100}{56/100} = \frac{10}{56} \approx 0.1786.$$

Iska matlab: Phone 2 Weak hone par, sirf 18% baar Phone 1 Strong nikalta hai.

Given that $Y = \text{Weak}$, the probability that $X = \text{Weak}$ will be:

$$p(X = \text{Weak} \mid Y = \text{Weak}) = \frac{46/100}{56/100} = \frac{46}{56} \approx 0.8214.$$

Iska matlab: Phone 2 Weak hone par, 82% baar Phone 1 bhi Weak hota hai — yahan bhi strong agreement.

Kyun? Ab har branch ka bacha confusion $H(X \mid Y = y)$ nikalte hain, surprise table se.

Step 3: $H(X \mid Y = y)$ **nikaalo**

$$H(X \mid Y = \text{Strong}) = \frac{35}{44}(0.2288) + \frac{9}{44}(1.5870) = 0.1820 + 0.3246 = 0.5066 \text{ nats.}$$
$$H(X \mid Y = \text{Weak}) = \frac{10}{56}(1.7228) + \frac{46}{56}(0.1967) = 0.3077 + 0.1615 = 0.4692 \text{ nats.}$$

Iska matlab: Dono branches me bacha confusion kaafi close hai (0.5066 aur 0.4692 nats), kyunki agreement dono taraf roughly 80% ke aas-paas hai.

Step 4: Average lo

$$H(X \mid Y) = \frac{44}{100}(0.5066) + \frac{56}{100}(0.4692) = 0.2229 + 0.2628 = 0.4857 \text{ nats.}$$

Kyun? Ab Phone 2 jaanne se Phone 1 ka confusion kitna kam hua, ye $I(X;Y)$ se dekhte hain.

Step 5: $I(X;Y)$ **nikaalo**

$$I(X;Y) = H(X) - H(X \mid Y) = 0.6881 - 0.4857 = 0.2025 \text{ nats.}$$

Iska matlab: Phone 1 ka signal-bucket jaanne se, Phone 2 ke baare me confusion 0.2025 nats kam hota hai — achha-khaasa reduction, kyunki dono ek hi tower se judi hain, bas chhota hardware-noise bachta hai.

Key Takeaway

- Independent (Jammu/Chennai weather): $I(X;Y) = 0$, circles alag.
- Identical (sensor aur uski copy): $I(X;Y) = H(X)$, circles poori tarah overlap.
- Partial (thermometers, commute-punctuality, cricket innings, mobile signal): $0 < I(X;Y) < H(X)$, circles kuch overlap — real life me yahi sabse common case hai.

5 Conditional Mutual Information: After Knowing a Third Variable Z

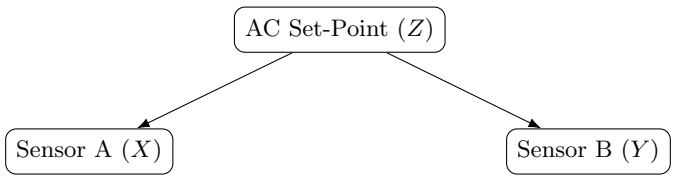
5.1 Definition

Ab ek teesra variable Z add karte hain — context ki tarah. Sawaal: Z pehle se pata hone par, Y jaanne se X ke baare me *aur* kitna milta hai?

$$I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z).$$

5.2 Common Cause Reduces Shared Information: AC Example

Story: Room me do thermometers hain — Sensor A aur Sensor B — aur ek AC jiska set-point Z hai (off ya on). Dono sensors AC ke hisaab se move karte hain: AC off hone par room garam hota hai (dono sensors High), AC on hone par room thanda hota hai (dono sensors Low). Z (AC-mode) ek **common cause** hai jo X, Y dono ko janam deta hai.



Z ke bina: X, Y dono high jab AC off hoti hai, dono low jab AC chalti hai — ek common cause (Z) unko correlated banata hai, isliye $I(X;Y)$ high hota hai.

Z =AC-mode pata hone ke baad: ab maloom hai AC kis mode me hai. X, Y ka jo correlation tha, wo mostly isi common AC-mode ki wajah se tha — Z ab wo explain kar chuka hai. Bacha hua shared information kam hota hai:

$$I(X;Y \mid Z) < I(X;Y).$$

Isko ab numbers ke saath verify karte hain.

Case: AC Mode, Two Sensors — Full Numeric Verification

Setup: Pichhle 100 din me se, 41 din AC off thi, 59 din AC on thi:

$$p(Z = \text{off}) = \frac{41}{100}, \qquad p(Z = \text{on}) = \frac{59}{100}.$$

Z pata hone par, X aur Y **conditionally independent** hain — dono sirf AC-mode par depend karte hain, ek-doosre par nahi (same room, same AC, par independent measurement-noise). AC-off me, 83/100 din Sensor High dikhata hai; AC-on me, 14/100 din High dikhata hai. Y ka bhi exact same pattern hai (same room, same AC):

$$p(X = \text{High} \mid Z = \text{off}) = \frac{83}{100}, \qquad p(X = \text{High} \mid Z = \text{on}) = \frac{14}{100}.$$

Kyun? Pehle $H(X \mid Z)$ chahiye — matlab, AC-mode (Z) pehle se pata hone par, Sensor A (X) me average kitna confusion bachta hai. Iske liye har AC-branch ka apna surprise table banate hain.

Step 1: $H(X \mid Z)$ nikaalo, pehle
AC-off branch ki surprise table:

$x \mid Z =$off	p	S
High	83/100	≈ 0.1863
Low	17/100	≈ 1.7720

$$H(X \mid Z = \text{off}) = \frac{83}{100}(0.1863) + \frac{17}{100}(1.7720) = 0.1546 + 0.3012 = 0.4559 \text{ nats.}$$

AC-on branch:

$x \mid Z =$on	p	S
High	14/100	≈ 1.9661
Low	86/100	≈ 0.1508

$$H(X \mid Z = \text{on}) = \frac{14}{100}(1.9661) + \frac{86}{100}(0.1508) = 0.2753 + 0.1297 = 0.4050 \text{ nats.}$$

$$H(X \mid Z) = \frac{41}{100}(0.4559) + \frac{59}{100}(0.4050) = 0.1869 + 0.2389 = 0.4258 \text{ nats.}$$

Iska matlab: AC-mode pata hone ke baad bhi, Sensor A me 0.4258 nats confusion bachta hai — AC akele poora nahi bata deta ki sensor High hoga ya Low, measurement-noise ki wajah se kuch confusion rehta hai.

Kyun? Ab $H(X \mid Y, Z)$ chahiye — matlab, AC-mode **aur** Sensor B dono pata hone par, Sensor A me kitna confusion bachta hai. Yahan setup ye bata chuka hai ki Z pata hone par X, Y conditionally independent hain, isliye Y ka extra pata hona, X ki conditional distribution ko bilkul nahi badalega.

Step 2: $H(X \mid Y, Z)$ nikaalo
Z pata hone par X, Y conditionally independent hain (setup se) — isliye Y ka pata hona, X ki conditional distribution ko *bilkul nahi* badalta, jab Z already pata hai:

$$H(X \mid Y, Z) = H(X \mid Z) = 0.4258 \text{ nats.}$$

Iska matlab: Y ka extra pata hona kuch nahi jodta, kyunki Z (AC-mode) hi X, Y dono ka asli karan hai — Z pata hote hi, Y sirf "noise" reh jaata hai jo X ko nahi batata.

Kyun? Ab $I(X; Y \mid Z)$ nikalte hain — matlab, Z pehle se pata hone par, Y jaanne se X ka confusion aur kitna kam hota hai.

Step 3: $I(X; Y \mid Z)$ nikaalo

$$I(X; Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z) = 0.4258 - 0.4258 = 0.$$

Iska matlab: AC-mode (Z) pata hone ke baad, Sensor A aur Sensor B ke beech ab koi connection nahi bachta — AC hi unka poora correlation explain kar deta tha.

Kyun? Ab yahi baat Z ke **bina** dekhte hain — matlab, $I(X;Y)$ (bina Z jaane) kitna bada tha, taaki Step 3 ke $I(X;Y \mid Z) = 0$ se compare kar sakein.

Step 4: $I(X;Y)$, bina Z ke, compare ke liye
Joint $p(X,Y)$ nikaalo, Z ko average karte hue:

$$\begin{aligned} p(\text{High},\text{High}) &= \frac{41}{100}\cdot\frac{83}{100}\cdot\frac{83}{100} + \frac{59}{100}\cdot\frac{14}{100}\cdot\frac{14}{100} \\ &= 0.2824 + 0.0116 \\ &= 0.2940. \end{aligned}$$

$$\begin{aligned} p(\text{High},\text{Low}) = p(\text{Low},\text{High}) &= \frac{41}{100}\cdot\frac{83}{100}\cdot\frac{17}{100} + \frac{59}{100}\cdot\frac{14}{100}\cdot\frac{86}{100} \\ &= 0.0579 + 0.0710 \\ &= 0.1289. \end{aligned}$$

$$\begin{aligned} p(\text{Low},\text{Low}) &= \frac{41}{100}\cdot\frac{17}{100}\cdot\frac{17}{100} + \frac{59}{100}\cdot\frac{86}{100}\cdot\frac{86}{100} \\ &= 0.0118 + 0.4364 \\ &= 0.4482. \end{aligned}$$

Iska matlab: Ye products isliye aaye kyunki, Z pata na hone par, X aur Y ka joint, har AC-branch ke andar dono ka (conditionally independent) product lekar, dono branches ka weighted-sum lena padta hai — matlab Z ko "average out" kiya gaya.

Marginal $p(X = \text{High}) = 0.2940 + 0.1289 = 0.4229$, isliye $H(X) = H(Y) \approx 0.6812$ nats. Joint entropy:

$$\begin{aligned} H(X,Y) &= 0.2940(1.2241) + 2(0.1289)(2.0488) + 0.4482(0.8025) \\ &= 0.3599 + 0.5282 + 0.3597 \\ &= 1.2477 \text{ nats}. \end{aligned}$$

$$\begin{aligned} I(X;Y) &= H(X) + H(Y) - H(X,Y) \\ &= 0.6812 + 0.6812 - 1.2477 \\ &= 0.1147 \text{ nats}. \end{aligned}$$

Iska matlab: Numbers confirm qualitative baat: $I(X;Y) \approx 0.1147$ (kuch shared information hai, Z ke bina), par $I(X;Y \mid Z) = 0$ (bilkul kuch bacha nahi, Z pata hone ke baad) — Z (AC mode) ne poora shared information explain kar diya.

5.3 Common Effect Increases Shared Information: Rain and Sprinkler

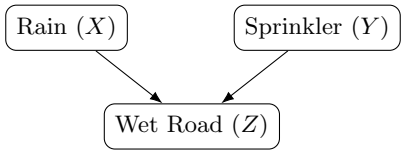
Story: X = aaj baarish hui (yes/no). Y = sprinkler chalaya gaya (yes/no). Z = road geeli hai (yes/no).

Z ke bina: baarish hona aur sprinkler chalana almost **unrelated** causes hain, ek-doosre se koi lena-dena nahi.

Z =road geeli pata hone ke baad: ab maloom hai road geeli hai, par wajah pata nahi. Agar ab pata chale Y =sprinkler chala, to ye baarish (X) ka explanation *hata* deta hai, road geeli hone ki ek wajah mil gayi, baarish ki zaroorat kam ho gayi. Yani Y jaanne se X ke baare me ab kuch pata chalta hai, jabki Z ke bina Y se X ka koi lena-dena nahi tha:

$$I(X;Y \mid Z) > I(X;Y).$$

Isko bhi numbers ke saath verify karte hain.



Case: Rain, Sprinkler, Wet Road — Full Numeric Verification

Setup: Pichhle 100 din ka record, X =baarish (1 =yes), Y =sprinkler (1 =yes):

X	Y	$p(x,y)$
0	0	44/100
0	1	11/100
1	0	25/100
1	1	20/100

$Z = 1$ agar baarish ya sprinkler me se koi bhi ho ($Z = X$ OR Y), warna $Z = 0$.

Kyun? Pehle dekhte hain Z ke **bina**, baarish aur sprinkler ke beech kitna shared information hai (baseline, taaki aage $I(X;Y \mid Z)$ se compare kar sakein).

Step 1: $I(X;Y)$ **nikaalo,** Z **ke bina**

$$p(X=1) = \frac{25}{100} + \frac{20}{100} = \frac{45}{100}, \qquad p(Y=1) = \frac{11}{100} + \frac{20}{100} = \frac{31}{100}.$$

Surprise table se, $H(X) \approx 0.6881$, $H(Y) \approx 0.6191$. Joint entropy:

$$\begin{aligned} H(X,Y) &= \frac{44}{100}(0.8210) + \frac{11}{100}(2.2073) + \frac{25}{100}(1.3863) + \frac{20}{100}(1.6094) \\ &= 0.3612 + 0.2428 + 0.3466 + 0.3219 \\ &= 1.2725 \text{ nats.} \end{aligned}$$

$$\begin{aligned} I(X;Y) &= H(X) + H(Y) - H(X,Y) \\ &= 0.6881 + 0.6191 - 1.2725 \\ &= 0.0347 \text{ nats.} \end{aligned}$$

Iska matlab: Ye bahut chhota hai — baarish aur sprinkler almost unrelated causes hain, jaisa setup me bataya.

Kyun? Ab Z =road-geeli ko condition karke dekhte hain. Pehle Z ki probability chahiye, phir har Z-branch ke andar $I(X;Y \mid Z=z)$ nikalenge.

Step 2: Z ki probability nikaalo

$$p(Z=1) = \frac{11}{100} + \frac{25}{100} + \frac{20}{100} = \frac{56}{100}, \qquad p(Z=0) = \frac{44}{100}.$$

Kyun? $Z=0$ branch ke andar sirf ek outcome bachta hai, isliye wahan koi confusion ho hi nahi sakta — dekhte hain.

Step 3: $Z=0$ branch ka $I(X;Y \mid Z=0)$ nikaalo

$Z=0$ ke andar sirf ek hi outcome hai $(X=0, Y=0)$ — koi confusion bachta hi nahi, isliye:

$$I(X;Y \mid Z=0) = 0.$$

Kyun? Ab $Z=1$ branch ke andar jaana hai. Pehle us branch ke andar (X,Y) ki conditional probabilities chahiye — matlab, $Z=1$ given hone par, teeno bache outcomes $(0,1),(1,0),(1,1)$ me se kaunsa kitna likely hai. Iske liye unki original joint probabilities ko $Z=1$ ke total $(56/100)$ se renormalize karna hoga.

Step 4: $Z=1$ branch ke andar conditional probabilities nikaalo

$Z=1$ ke andar teen outcomes bachte hain — $(0,1),(1,0),(1,1)$, original probabilities $11/100, 25/100, 20/100$, total $56/100$ me se renormalize:

Given that $Z=1$, the probability that $(X,Y)=(0,1)$ will be:

$$p(0,1 \mid Z=1) = \frac{11/100}{56/100} = \frac{11}{56} \approx 0.1964.$$

Iska matlab: 56 din me se jab road geeli thi, 11 din sirf sprinkler ki wajah se (baarish nahi hui thi).

Given that $Z=1$, the probability that $(X,Y)=(1,0)$ will be:

$$p(1,0 \mid Z=1) = \frac{25/100}{56/100} = \frac{25}{56} \approx 0.4464.$$

Iska matlab: 56 din me se 25 din sirf baarish ki wajah se road geeli thi (sprinkler nahi chala).

Given that $Z=1$, the probability that $(X,Y)=(1,1)$ will be:

$$p(1,1 \mid Z=1) = \frac{20/100}{56/100} = \frac{20}{56} \approx 0.3571.$$

Iska matlab: Baaki 20 din dono (baarish aur sprinkler) ek saath hue the.

Kyun? Ab $H(X \mid Z=1)$ chahiye — matlab, road geeli (aur $Z=1$) pata hone par, baarish (X) me kitna confusion bachta hai. Iske liye pehle marginal $p(X \mid Z=1)$ nikaalenge, upar wali teen conditionals se jodkar.

Step 5: $H(X \mid Z=1)$ nikaalo

Marginal: $p(X=1 \mid Z=1) = p(1,0 \mid Z=1) + p(1,1 \mid Z=1) = 25/56 + 20/56 = 45/56 \approx 0.8036$, $p(X=0 \mid Z=1) = p(0,1 \mid Z=1) = 11/56 \approx 0.1964$.

$$H(X \mid Z=1) = \frac{45}{56}(0.2187) + \frac{11}{56}(1.6275) = 0.1758 + 0.3197 = 0.4954 \text{ nats.}$$

Iska matlab: Road geeli hone ke baad bhi, baarish hui ya nahi, isme 0.4954 nats confusion bachta hai — road geeli hona akela nahi batata ki wajah baarish thi ya sprinkler.

Kyun? Ab $H(X \mid Y, Z=1)$ chahiye — matlab, road geeli **aur** sprinkler ki haalat, dono pata hone par, baarish me kitna confusion bachta hai. Iske liye $Z=1$ ke andar, $Y=0$ aur $Y=1$ ke do sub-branches dekhne honge.

Step 6: $H(X \mid Y, Z=1)$ **nikaalo**

Agar $Y=0$ (aur $Z=1$): sirf $(1,0)$ possible — $X=1$ guaranteed, confusion 0.

Agar $Y=1$ (aur $Z=1$): $(0,1)$ aur $(1,1)$ possible, conditional probabilities $11/56$ aur $20/56$ hain, jinka sum $31/56 = p(Y=1 \mid Z=1)$ hai. In dono ko ab $Y=1, Z=1$ ke andar renormalize karte hain:

Given that $Y=1$ and $Z=1$, the probability that $X=0$ will be:

$$p(X=0 \mid Y=1, Z=1) = \frac{11/56}{31/56} = \frac{11}{31} \approx 0.3548.$$

Iska matlab: Jab sprinkler chala tha aur road geeli thi, 31 cases me se 11 cases me baarish nahi hui thi — yani sprinkler akele bhi road geeli kar sakta hai.

Given that $Y=1$ and $Z=1$, the probability that $X=1$ will be:

$$p(X=1 \mid Y=1, Z=1) = \frac{20/56}{31/56} = \frac{20}{31} \approx 0.6452.$$

Iska matlab: Baaki 65% cases me, sprinkler chalne ke saath baarish bhi hui thi.

$$H(X \mid Y=1, Z=1) = \frac{11}{31}(1.0361) + \frac{20}{31}(0.4383) = 0.3676 + 0.2828 = 0.6504 \text{ nats.}$$

Iska matlab: $Y=1, Z=1$ branch ke andar, baarish ka confusion 0.6504 nats hai — $Y=0$ branch ke 0 confusion se kaafi zyada, kyunki sprinkler chalna baarish ke baare me kuch nahi batata.

$p(Y=0 \mid Z=1) = 25/56 \approx 0.4464$, $p(Y=1 \mid Z=1) = 31/56 \approx 0.5536$, isliye average:

$$H(X \mid Y, Z=1) = 0.4464(0) + 0.5536(0.6504) = 0.3600 \text{ nats.}$$

Kyun? Ab $I(X;Y \mid Z=1)$ nikalte hain — $Z=1$ branch ke andar, sprinkler jaanne se baarish ka confusion kitna kam hua.

Step 7: $I(X;Y \mid Z=1)$ **nikaalo**

$$I(X;Y \mid Z=1) = H(X \mid Z=1) - H(X \mid Y, Z=1) = 0.4954 - 0.3600 = 0.1354 \text{ nats.}$$

Iska matlab: $Z = 1$ (road geeli) branch ke andar, sprinkler ki haalat jaanne se baarish ka confusion 0.1354 nats kam hota hai — explaining-away ki wajah se, $Z = 0$ branch ke 0 se kaafi zyada.

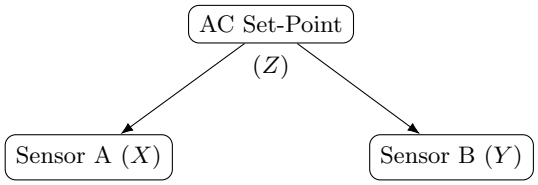
Kyun? Ab dono branches ($Z = 0$ aur $Z = 1$) ka weighted average lekar overall $I(X;Y \mid Z)$ nikalte hain.

Step 8: Dono branches ka weighted average lo

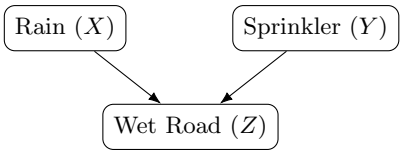
$$I(X;Y \mid Z) = p(Z=0) \cdot 0 + p(Z=1) \cdot 0.1354 = \frac{44}{100}(0) + \frac{56}{100}(0.1354) = 0.0758 \text{ nats.}$$

Iska matlab: Numbers confirm qualitative baat: $I(X;Y) \approx 0.0347$ (bahut chhota, Z ke bina), par $I(X;Y \mid Z) \approx 0.0758$ (Z pata hone ke baad) — Z (road geeli) ne in dono ke beech connection **bana** diya, explaining-away ki wajah se. Yani $I(X;Y \mid Z) > I(X;Y)$, numbers se confirmed.

5.4 Picture: Common Cause Versus Common Effect



Common **cause**: Z explains away shared info



Common **effect**: Z creates shared info

Key Takeaway

- Common **cause** (Z dono ko janam deta hai, jaise AC): Z jaanne se shared information **kam** hota hai.
- Common **effect** (Z dono se banta hai, jaise wet road): Z jaanne se shared information **badh** sakta hai.

6 O-information: Redundant Versus Synergistic

6.1 Definition

$$\Omega = I(X;Y) - I(X;Y \mid Z).$$

Sign ka Matlab

- $\Omega > 0 \implies$ redundant (Z se shared info kam hua).
- $\Omega < 0 \implies$ synergistic (Z se shared info badha).

6.2 Redundant Example: AC and Two Sensors

Kyun? Chapter 5 ke AC-mode example me, $I(X;Y)$ aur $I(X;Y \mid Z)$ dono already nikal chuke hain. Ab bas Ω ki definition me daal kar sign dekhte hain.

Chapter 5 ka AC-mode numeric example wapas lo: $I(X;Y) \approx 0.1147$ nats, $I(X;Y \mid Z) = 0$. Isliye:

$$\Omega = I(X;Y) - I(X;Y \mid Z) = 0.1147 - 0 = 0.1147 > 0.$$

Iska matlab: Ye **redundant** hai — Z (AC mode) common cause hai, dono sensors ki poori 0.1147 nats shared information ko explain kar deta hai, kuch bachta nahi.

6.3 Synergistic Example: Fire Alarm

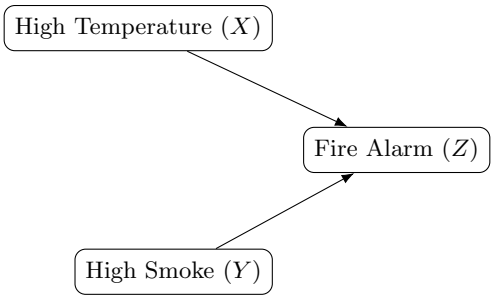
Story: X = high temperature detect hua (yes/no). Y = high smoke detect hua (yes/no). Z = fire alarm trigger hua (yes/no).

Individually, dono signals kamzor hain: Akela high temperature, AC fail hone se bhi ho sakta hai, zaroori nahi fire ho. Isliye $I(X;Z)$ chhota hai. Akela high smoke, cooking se bhi ho sakta hai, zaroori nahi fire ho. Isliye $I(Y;Z)$ bhi chhota hai. Temperature aur smoke khud bhi normally independent hain: $I(X;Y) \approx 0$.

Saath me, dono signals strong ban jaate hain: Agar **dono** saath me ho — high temperature *aur* high smoke — to fire almost certain hai. Akela koi ek signal itna confident nahi karta. Matlab: Y (smoke) jaanne se X (temperature) ke baare me zyada nahi pata chalta jab tak Z (fire) pata na ho. Par Z pata hone ke baad, X aur Y ka connection **expose** ho jaata hai — dono saath milkar fire confirm karte hain:

$$I(X;Y \mid Z) > I(X;Y) \quad \implies \quad \Omega = I(X;Y) - I(X;Y \mid Z) < 0.$$

Ye **synergistic** hai — do kamzor signals (temperature, smoke) milkar ek strong signal (fire) dete hain, jo akele kisi se nahi milta. Isko numbers ke saath bhi verify karte hain, ek simple design ke saath: real fire-alarm systems me false alarms kam karne ke liye **AND logic** common hai — alarm sirf tab trigger hota hai jab **dono** sensors high dikhayein.



Case: AND-Logic Fire Alarm — Full Numeric Verification

Setup: Pichhle 100 monitoring-din me se, 37 din high-temperature event hua, 23 din high-smoke event hua, dono **independent**:

$$p(X=1)=\frac{37}{100}, \qquad p(Y=1)=\frac{23}{100}.$$

Z = fire alarm, AND-logic se: $Z=1$ sirf jab $X=1$ **aur** $Y=1$ dono ho, warna $Z=0$.

Kyun? Pehle baseline $I(X;Y)$ chahiye, Z ke bina — kyunki setup me hi X, Y independent diye gaye hain, ye seedha nikal jaayega.

Step 1: $I(X;Y)$ nikaalo, Z ke bina

X, Y independent diye gaye hain (setup se), isliye seedha:

$$I(X;Y)=0.$$

Kyun? Ab Z ko define karne ke liye joint $p(x,y)$ chahiye — X, Y independent hain, isliye joint seedha dono fractions ka product hoga.

Step 2: Joint $p(x,y)$ nikaalo, independence se

$$p(1,1)=\frac{37}{100}\cdot\frac{23}{100}=\frac{851}{10000}=0.0851. \qquad p(1,0)=\frac{37}{100}\cdot\frac{77}{100}=\frac{2849}{10000}=0.2849.$$
$$p(0,1)=\frac{63}{100}\cdot\frac{23}{100}=\frac{1449}{10000}=0.1449. \qquad p(0,0)=\frac{63}{100}\cdot\frac{77}{100}=\frac{4851}{10000}=0.4851.$$

Kyun? Ab Z (fire alarm) ki probability chahiye. AND-logic se, $Z=1$ sirf $(X,Y)=(1,1)$ wale case me hota hai.

Step 3: Z ki probability nikaalo

$$p(Z=1)=p(1,1)=\frac{851}{10000}=0.0851, \qquad p(Z=0)=\frac{9149}{10000}=0.9149.$$

Kyun? Ab $Z=0$ branch ke andar jaana hai — matlab, jab fire alarm trigger **nahi** hua, tab (X,Y) ki conditional probabilities kya hain. Iske liye teen bache outcomes $(1,0),(0,1),(0,0)$ ko $Z=0$ ke total se renormalize karte hain.

Step 4: $Z=0$ branch ke andar $p(x,y \mid Z=0)$ nikaalo

$Z=0$ ke andar teen combinations bachte hain — $(1,0),(0,1),(0,0)$, original probabilities $2849/10000, 1449/10000, 4851/10000$, total $9149/10000$ me se renormalize:

Given that $Z = 0$, the probability that $(X, Y) = (1, 0)$ will be:

$$p(1, 0 \mid Z = 0) = \frac{2849/10000}{9149/10000} = \frac{2849}{9149} \approx 0.3114.$$

Iska matlab: Jab alarm nahi baja, 31% cases me sirf temperature high tha (smoke nahi).

Given that $Z = 0$, the probability that $(X, Y) = (0, 1)$ will be:

$$p(0, 1 \mid Z = 0) = \frac{1449/10000}{9149/10000} = \frac{1449}{9149} \approx 0.1584.$$

Iska matlab: 16% cases me sirf smoke high tha (temperature nahi).

Given that $Z = 0$, the probability that $(X, Y) = (0, 0)$ will be:

$$p(0, 0 \mid Z = 0) = \frac{4851/10000}{9149/10000} = \frac{4851}{9149} \approx 0.5302.$$

Iska matlab: Baaki 53% cases me dono hi normal the — yahi sabse common situation hai.

Kyun? Ab $H(X \mid Z = 0)$ chahiye — matlab, alarm na bajne ki haalat me, temperature-sensor (X) me kitna confusion bachta hai. Iske liye marginal $p(X \mid Z = 0)$ chahiye, upar wali conditionals se.

Step 5: $H(X \mid Z = 0)$ **nikaalo**
Marginal: $p(X = 1 \mid Z = 0) = 2849/9149 \approx 0.3114$, $p(X = 0 \mid Z = 0) = 6300/9149 \approx 0.6886$.

$$H(X \mid Z = 0) = 0.3114(1.1667) + 0.6886(0.3731) = 0.3633 + 0.2569 = 0.6202 \text{ nats.}$$

Iska matlab: Alarm na bajne par bhi, temperature high tha ya nahi, isme 0.6202 nats confusion bachta hai.

Kyun? Ab $H(X \mid Y, Z = 0)$ chahiye — matlab, alarm na bajne **aur** smoke-sensor ki haalat dono pata hone par, temperature-sensor ka confusion kitna bachta hai. Iske liye $Z = 0$ ke andar $Y = 1$ aur $Y = 0$ sub-branches dekhte hain.

Step 6: $H(X \mid Y, Z = 0)$ **nikaalo**
Agar $Y = 1$ (aur $Z = 0$): sirf $(0, 1)$ possible — $X = 0$ guaranteed, confusion 0.
Agar $Y = 0$ (aur $Z = 0$): $(1, 0)$ aur $(0, 0)$ possible, renormalize:
Given that $Y = 0$ and $Z = 0$, the probability that $X = 1$ will be:

$$p(X = 1 \mid Y = 0, Z = 0) = \frac{2849/9149}{7700/9149} = \frac{2849}{7700} \approx 0.3700.$$

Given that $Y = 0$ and $Z = 0$, the probability that $X = 0$ will be:

$$p(X = 0 \mid Y = 0, Z = 0) = \frac{4851}{7700} \approx 0.6300.$$

Iska matlab: Jab smoke bhi nahi tha aur alarm bhi nahi baja, 37% baar temperature phir bhi high tha — AC fail jaisi akeli wajahon se.

$H(X \mid Y = 0, Z = 0) = 0.3700(0.9943)+0.6300(0.4620) = 0.3679+0.2911 = 0.6590$ nats.

$p(Y = 0 \mid Z = 0) \approx 0.8416$, $p(Y = 1 \mid Z = 0) \approx 0.1584$:

$$H(X \mid Y, Z = 0) = 0.8416(0.6590) + 0.1584(0) = 0.5546 \text{ nats.}$$

Kyun? Ab $I(X;Y \mid Z = 0)$ nikalte hain — $Z = 0$ branch ke andar, smoke-sensor jaanne se temperature-sensor ka confusion kitna kam hua.

Step 7: $I(X;Y \mid Z = 0)$ **nikaalo**

$$I(X;Y \mid Z = 0) = H(X \mid Z = 0) - H(X \mid Y, Z = 0) = 0.6202 - 0.5546 = 0.0656 \text{ nats.}$$

Iska matlab: Alarm na bajne ki haalat me, smoke-sensor jaanne se temperature-sensor ka confusion 0.0656 nats kam hota hai — chhota, par zero nahi, kyunki AND-logic ke andar dono ab bhi thoda link rakhte hain.

Kyun? $Z = 1$ branch ke andar sirf ek hi combination bachta hai, isliye wahan koi confusion ho hi nahi sakta — dekhte hain.

Step 8: $Z = 1$ **branch nikaalo**
$Z = 1$ ke andar sirf ek hi combination hai $(X = 1, Y = 1)$, koi confusion bachta hi nahi:

$$I(X;Y \mid Z = 1) = 0.$$

Kyun? Ab dono branches ka weighted average lekar overall $I(X;Y \mid Z)$ milega.

Step 9: Dono branches ka weighted average lo, $I(X;Y \mid Z)$ **milega**

$$I(X;Y \mid Z) = p(Z = 0) \cdot 0.0656 + p(Z = 1) \cdot 0 = 0.9149(0.0656) = 0.0600 \text{ nats.}$$

Kyun? Ab Ω nikalte hain — $I(X;Y)$ aur $I(X;Y \mid Z)$ ka farak, taaki sign se redundant/synergistic decide ho.

Step 10: Ω **nikaalo**

$$\Omega = I(X;Y) - I(X;Y \mid Z) = 0 - 0.0600 = -0.0600.$$

Iska matlab: Ω negative hai — confirm hota hai ki ye **synergistic** hai. X, Y akele (Z ke bina) bilkul unrelated hain ($I(X;Y) = 0$), par Z (fire-alarm trigger) pata hone ke baad X, Y ke beech 0.0600 nats ka connection expose ho jaata hai, jo pehle dikh hi nahi raha tha.

Key Takeaway

- $\Omega = I(X;Y) - I(X;Y \mid Z)$. Positive sign = redundant, negative sign = synergistic.
- Redundant: common cause (AC) explains away shared info between sensors.
- Synergistic: individually-weak signals (temperature, smoke) combine into a strong one (fire) only once Z is known.

7 Why Logarithm: The Additive Property, Revisited

7.1 Multiply Versus Add

Chapter 1 me informally dekha tha ki do independent coins ka surprise, ek coin ke surprise se double hota hai. Ab isko thoda formally dekhte hain.

Do independent events A, B ke liye, unka saath hona (A and B) ki probability **multiply** hoti hai:

$$P(A \text{ and } B) = P(A) \times P(B).$$

Lekin information (ya surprise) **add** honi chahiye — do independent cheezon ke baare me jitni information milti hai, wo alag-alag mili information ka **sum** honi chahiye, product nahi.

Logarithm hi ye conversion karta hai:

$$\ln(a \times b) = \ln a + \ln b.$$

Isliye surprise ko $-\ln p$ define kiya gaya, seedha p nahi — taaki multiply-hone-wali probabilities, add-hone-wale surprise me convert ho jaayein.

7.2 Numeric Check

Case: Two Independent Rooms

Setup: Room 1 ka temperature-bucket (X_1, High/Low) aur Room 2 ka temperature-bucket (X_2, High/Low), dono rooms unrelated, dono $p(\text{High}) = p(\text{Low}) = 1/2$.

Step 1: Ek room ka surprise

$$S(X_1 = \text{High}) = -\ln \frac{1}{2} = \ln 2 \approx 0.693.$$

Kyun? Ab dono rooms ka joint surprise chahiye, taaki dekh sakein ki ye ek room ke surprise ka exactly double aata hai ya nahi — additivity ka check.

Step 2: Dono rooms ka joint surprise

Independent hain, isliye joint probability multiply hogi:

$$p(X_1 = \text{High}, X_2 = \text{High}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Joint surprise:

$$S = -\ln \frac{1}{4} = \ln 4 \approx 1.386.$$

Step 3: Additivity verify karo

$$\ln 4 = \ln 2 + \ln 2.$$

Iska matlab: Joint surprise, individual surprises ka **sum** hai — bilkul jaisa chahiye tha.

7.3 Why Not Just Use Probability

Agar surprise ko seedha probability se define karte, ye additivity *nahi* milti.

Analogy: Coin par bet lagate ho, Heads par jeetne ka chance 1/2 hai. Do independent coins par bet lagao aur dono Heads chahiye, to jeetne ka chance $1/2 \times 1/2 = 1/4$ hai — chance multiply ho gaya, add nahi hua.

Agar "surprise" ko probability p hi maan lete, to do coins ka surprise 1/4 hota (sirf multiply), jabki do independent surprises ka sum $1/2 + 1/2 = 1$ hona chahiye tha. Ye match nahi karta.

Logarithm hi ye theek karta hai:

$$-\ln\left(\frac{1}{4}\right) = -\ln\left(\frac{1}{2} \times \frac{1}{2}\right) = -\ln\frac{1}{2} - \ln\frac{1}{2} = \ln 2 + \ln 2 = \ln 4 \approx 1.386.$$

Multiply probability ko, add surprise me convert kar deta hai.

7.4 So: $\ln 2$ Versus $1/2$, Once More

- $1/2$ — ek outcome ki **probability**. Multiply-hone-wali quantity.
- $-\ln(1/2) = \ln 2$ — us outcome ka **surprise**. Add-hone-wali quantity.
- $\ln 2 \approx 0.693$ (fair coin ka entropy) — surprise ka **average**, saare outcomes par.

Teeno alag hain, aur teeno apni apni jagah zaroori hain.

Key Takeaway

- Independent events ki probabilities multiply hoti hain; unki information add honi chahiye.
- Logarithm multiply ko add me convert karta hai: $\ln(ab) = \ln a + \ln b$.
- Isliye surprise $-\ln p$ hai, probability p nahi — aur entropy, is surprise ka average.

Bridge: Connecting This Document to the Higher-MI Walkthrough

Is document ka maqsad tha: Higher-MI walkthrough padhne se **pehle**, basic building-blocks (entropy, conditional entropy, mutual information, conditional mutual information, O-information) ko real-life examples se intuitive banana. Ab Higher-MI walkthrough wapas padhte waqt, ye mapping yaad rakhna:

- Higher-MI me $H(X)$, $H(X \mid Y)$ dikhe — yaad karo Chapter 1–2 ka band-dabba aur thermometer example.
- Higher-MI me $I(X;Y)$ dikhe, aur overlap-circle diagram dikhe — yaad karo Chapter 3–4 ka "reduction, not final value" aur teen cases (independent/identical/partial).

- Higher-MI me $I(X;Y \mid Z)$ dikhe, common-cause ya common-effect ki baat ho — yaad karo Chapter 5 ka AC-mode aur rain-sprinkler example.
- Higher-MI me Ω (O-information) dikhe, redundant ya synergistic ki baat ho — yaad karo Chapter 6 ka AC (redundant) aur fire-alarm (synergistic) example.
- Higher-MI me $\ln 2 \approx 0.693$ dikhe aur confusion ho ki ye 0.5 kyun nahi — yaad karo Chapter 1 aur Chapter 7 ka additivity-wala explanation.

Agar Higher-MI walkthrough me kahin formula clear lage par intuition na bane, wapas isi document ke corresponding chapter par aa jao — formula wahi rahega, bas example badal jaayega.

Appendix: Common Doubts, Answered

Ye appendix un doubts ko seedhe jawab deta hai jo Higher-MI walkthrough padhte waqt aksar uthte hain. Har jawab wahi example reuse karta hai jo upar ke chapters me already bana hua hai.

Doubt 1: Redundancy ke example me $I(X;Y)$ zero kyun nahi hai?

Agar $X = Y$ (identical case, Chapter 3), to $I(X;Y) = H(X)$, zero nahi — ye **maximum** hai, zero nahi. Redundancy ka matlab ye nahi ki $I(X;Y) = 0$; redundancy ka matlab hai $I(X;Y \mid Z)$, $I(X;Y)$ se kam ya barabar ho jaata hai, matlab Z us shared information ko explain kar deta hai. Chapter 5–6 ke AC numeric case me $I(X;Y) \approx 0.1147$ hai, $I(X;Y \mid Z) = 0$ hai — $I(X;Y)$ khud zero nahi tha, par Z jaanne ke baad woh poora explain ho gaya.

Doubt 2: $I(X;Y)$ ka intuitive matlab kya hai?

$I(X;Y)$ batata hai: Y jaanne se X ka confusion **kitna kam hua**. Picture: do overlapping circles (Chapter 3), overlap wala hissa $I(X;Y)$ hai. Teen cases hote hain (Chapter 4): independent (overlap zero), identical (poora overlap), partial (kuch overlap) — real life me partial sabse common hai.

Doubt 3: Agar $X = Y$, to reduction 0 hona chahiye na, $\ln 2$ kaise?

Yahi Chapter 3 ka core confusion tha, Section 3.6 ("Resolving the Confusion") me seedha address kiya gaya. Do alag sawaal hain: "Y jaanne ke baad kitna confusion *bacha*?" (jawab: $H(X \mid Y) = 0$, jab $X = Y$) aur "Y jaanne se confusion kitna *kam* hua?" (jawab: $I(X;Y) = \ln 2$, jab $X = Y$, kyunki poora confusion hi kam ho gaya). Dono sahi hain — dono alag quantities hain, alag sawaalon ke jawab.

Doubt 4: "Confusion" ko imagine karne me problem ho rahi hai

Chapter 1 ka band-dabba wala tasveer yahi fix karne ke liye hai: dabbe ke andar coin hai, tumhe nahi pata kya hai — yahi confusion hai. Dabba khulte hi (ya Y jaanne se), confusion kam ya khatam ho jaata hai. Poore document me jahan bhi "confusion" aaye, isi dabbe ko yaad karo.

Doubt 5: 50-50 chance ka confusion 0.5 jaisa hona chahiye, 0.693 kyun?

Chapter 1, Section 1.4 ("Why 0.693, Not 0.5") aur Chapter 7 ("Why Logarithm") dono iska jawab hain. 0.5 ek **probability** hai; 0.693 us probability ka **surprise** ($-\ln 0.5$), average kiya hua. Dono alag quantities hain — probability multiply hoti hai, surprise add honi chahiye, isliye surprise ko $-\ln p$ se define kiya gaya, seedha p nahi.

Doubt 6: Entropy ka formula $-\sum p(x)\ln p(x)$ hai, par practically ye kaise use karte hain?

Formula ka matlab hai: har outcome ka surprise ($-\ln p(x)$) nikaalo, phir usi outcome ki probability se weight karke average lo. Is poore document me jis bhi numeric Case me $H(X)$ nikala gaya (Chapter 1 ka commute-mode, Chapter 2–6 ke sensor/cricket/mobile/AC/fire-alarm examples), wahi 2-3 step process repeat hua hai: Step A — har outcome ka surprise; Step B — weighted average. Koi shortcut nahi, har baar process.

Doubt 7: Nats aur bits me kya farak hai?

Chapter 1 ke "Bits Versus Nats" note me bataya gaya: ln use karne se unit nats hoti hai, $\log_2$ karne se bits. Ye document hamesha ln (nats) use karta hai, taaki Higher-MI walkthrough se match kare. Agar kisi EC course me bits chahiye, sirf $\log_2 x = \ln x/\ln 2$ se convert kar lo — concept (reduction, confusion, redundancy, synergy) bilkul same rehta hai, sirf number ka scale badalta hai.

Doubt 8: Har example me fraction /100 kyun likha gaya hai, seedha decimal kyun nahi?

Taaki probability ka source dikhta rahe — jaise 34/100 ka matlab hai "100 din me se 34 din". Beech ke calculation me fraction form rakhne se, har step me ye yaad rehta hai ki number kahan se aaya. Decimal sirf jab surprise nikalna ho ($-\ln$ lagana ho) ya final sum karna ho, tab use hota hai — beech me nahi.

Summary Table: Saare Numeric Examples Ek Jagah

Example (Chapter)	$H(X)$	$H(X \mid Y)$	$I(X;Y)$
Fair coin (Ch. 1)	0.693	—	—
Commute mode, Bike/Car (Ch. 1)	0.6914	—	—
Two home thermometers (Ch. 2–3)	0.6859	0.4357	0.2502
Identical coins, $Y = X$ (Ch. 3)	0.693	0	0.693
Commute mode vs punctuality (Ch. 3)	0.6914	0.5834	0.1080
Cricket innings (Ch. 4)	0.6881	0.3634	0.3247
Mobile signal, two phones (Ch. 4)	0.6881	0.4857	0.2025

Iska matlab: Ek pattern dikhta hai: jab X, Y bilkul unrelated hote hain, $I(X;Y) \to 0$; jab X, Y identical hote hain, $I(X;Y) \to H(X)$ (maximum); baaki sab real-life examples beech mein kahin partial reduction dete hain — yahi is poore document ka core idea hai.